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# Monitoring Flood Inundation Dynamics From Space

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**Key Points:**

- Sensors capable of observing floods doubled from 2015 to 2020; collectively they capture every major event despite individual gaps
- Synergistic fusion of frequent operational data and precise scientific imagery is critical to achieve continuous flood monitoring

**Supporting Information:**

Supporting Information may be found in the online version of this article.

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**Abstract** With the increasing intensity and frequency of flood events worldwide, the need for accurate and timely inundation mapping has never been more critical. Large-scale flood extent estimations are vital for coordinating effective disaster response, facilitating recovery, and building future resilience. Traditional ground-based and aerial monitoring methods are often impractical during major floods, limited by cost, safety, and their inability to capture the full scope of an event. Satellite-based remote sensing provides the necessary large-scale perspective with a unique vantage point to monitor extreme inundation events. This review assesses the potential of public satellite sensors to capture flood events using a novel analysis of the Dartmouth Flood Observatory (DFO) global flood database. Our analysis quantifies the major performance gaps between these sensors, demonstrating that no single instrument is sufficient for complete and continuous flood monitoring. Passive microwave radiometers are capable of capturing >95% of flood events, albeit at a coarse spatial resolution that may be unsuitable for detailed mapping or local risk assessment. In contrast, popular multispectral sensors such as Landsat and Sentinel-2 capture no more than 30% of flood events. The number of sensors capable of capturing flood events doubled between 2015 and 2020, signaling immense potential for multi-sensor integration. We examine how combining observations from multiple sensors can improve temporal coverage of flood events, however noting that temporal sampling along does not guarantee successful flood detection and how the rapid, dynamic nature of floods compounds the challenges inherent to satellite-based monitoring.

**Plain Language Summary** Globally, flooding is a marked issue that is growing in importance with the impacts of climate change. Flooding can occur at large scales, making it advantageous to use space-based remote sensing technology to monitor these hazards. Numerous public satellites are available, but their capabilities vary widely. We provide a data-driven analysis of thousands of global flood events to objectively measure and compare the performance of the different sensors available. Our analysis shows that combining data from different satellites increases the likelihood of capturing flood events. While no single sensor can reliably observe all floods, using multiple sensors together improves temporal coverage, though challenges remain in translating observations into accurate flood maps. The review also highlights the difficulties of using satellite data effectively due to the rapid and dynamic nature of floods, making a multi-sensor approach logical.

## 1. Introduction

Water is a valuable resource that can have significant consequences when in excess or in deficit, making water monitoring and management an important task. Effective water management hinges on accurate monitoring, particularly of inland surface water distribution across space and time. This information is essential for procedures such as accurate water resource assessment, environmental water management, and flood risk assessment (Mueller et al., 2016; Sheffield et al., 2018). Surface water exhibits diverse behaviors, ranging from relatively static bodies like lakes and reservoirs with seasonal fluctuations, to highly dynamic systems like floods.

Humans have historically settled near fertile lowlands and floodplains because of their importance for food production and other ecosystem services (water, recreation, and cultural significance). While early settlements favored areas with less frequent flooding, advances in water management and flood control have enabled expansion into flood-prone zones. More than 50% of the global population is estimated to reside within 3 km of a terrestrial water body, with many populations dependent on the seasonal inundation patterns and increasingly vulnerable to extreme flood events (Kummu et al., 2011; Tellman et al., 2021). Seasonal inundation is vital in many regions for agriculture, with 24% of the world's croplands in seasonally inundated floodplains (Dryden

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et al., 2021). Seasonal inundation aids soil health, refreshes nutrients, and is essential for certain crops, such as rice (Triet et al., 2020). However, prolonged, and excess flooding can cause crops to fail, potentially disrupting national economies and regional or even global food supplies (Maitah et al., 2020). Globally, disasters resulting from extreme inundation are both widespread and destructive, accounting for 45% of all reported disasters globally between 1970 and 2021, with economic losses attributed to floods nearly doubling between 2000–2009 and 2010–2019 (World Meteorological Organization, 2023). This increasing burden is driven by the rising frequency, intensity, and duration of flooding, with the extended duration of inundation being of particular concern for flood management. Given the complex and widespread nature of flood risks, particularly in cross-border river basins, Earth Observation (EO) has tremendous potential for monitoring and mitigation (UNEP-DHI & UNEP, 2016).

Critically, detection and monitoring of extreme inundation are necessary for effective mitigation and management. The varying spatial and temporal scales at which inundation progresses require consistent monitoring. However, it is difficult to guarantee a reliable data stream with field-based methods due to access difficulties and limited resources. Monitoring inundation through numerical modeling presents challenges for evaluating large-scale events, primarily due to uncertainties in global flood models, limited scalability, and inconsistent in situ calibration data (Bates et al., 2014; Fleischmann et al., 2019; Trigg et al., 2016). Remote sensing technologies enable inundation monitoring across different spatial and temporal scales, bypassing these limitations to yield consistent observations. The frequent and multi-scale monitoring capabilities of satellite-based remote sensing technologies make them highly relevant for large-scale inundation monitoring. This has long motivated researchers to evaluate the many possibilities for leveraging satellites for water surface monitoring with various types of sensors. Early efforts to leverage satellite technology for surface water monitoring, such as Bhavsar's (1984) evaluation of aerial photography and early Landsat imagery, laid the foundation for understanding the potential of remote sensing in hydrology. The proliferation of Synthetic Aperture Radar (SAR) satellites in the 1990s, such as ERS-1 and AIRSAR, marked notable advancements for flood mapping. Smith (1997) highlighted the distinct advantages of SAR for monitoring river inundation area, stage, and discharge, particularly its ability to penetrate clouds and acquire data day or night. This capability addressed a key limitation of optical sensors to enable flood detection and mapping even under the often-clouded conditions that are prevalent during flood events. The specific application of remote sensing for effective flood hazard assessment is explored in the work of Brakenridge et al. (2003) and Brakenridge and Anderson (2006), who demonstrate the potential of high temporal frequency satellites, such as the Moderate Resolution Imaging Spectroradiometer (MODIS), to actively monitor large-scale flood dynamics, however acknowledging the limitations posed by cloud coverage.

As the field matured, reviews began to synthesize the capabilities of optical, SAR and altimetry-based methods for flood mapping (Albertini et al., 2022; Alsdorf et al., 2007; Huang et al., 2018; International Altimetry Team, 2021; Meng et al., 2024; Schumann & Moller, 2015). These reviews enhanced understanding and application of remote sensing for flood purposes, but they also highlighted a critical problem: the continued emphasis on a siloed, single-sensor approach to flood detection and monitoring. This is despite the clear consensus in the literature that every sensor has inherent limitations. Consequently, successfully monitoring surface water at reasonably high temporal and spatial resolutions with open-source data necessitates a multi-sensor, multi-technique approach (Albertini et al., 2022; Huang et al., 2018; Meng et al., 2024; Munawar et al., 2021). Finding the right combination of sensors and methodologies is a daunting task which first requires selection of the appropriate sensors, thus requiring intimate knowledge of how the various types of sensors detect surface water from space, in addition to their core characteristics such as operational status, temporal resolution, and spatial resolution. However, a comprehensive review detailing these technical components and how they may be considered for optimal data fusion for inundation monitoring purposes is lacking. Furthermore, a comprehensive, quantitative assessment of the collective performance of public space-based EO sensors is missing. This review addresses these gaps, demonstrating that while no single sensor can reliably monitor all flood events, the strategic integration of multiple complementary sensors can achieve near-complete coverage of significant flood events. This review therefore aims to address the following questions: (a) What are the key capabilities, weaknesses, and operational differences among the space-based public EO sensors available for flood monitoring? (b) Given these capabilities, to what extent do these sensors successfully capture significant floods, as quantified by the proportion of the events in the Dartmouth Flood Observatory (DFO) database for which corresponding, valid, sensor data exists? Furthermore (c) How can a synergistic, multi-sensor framework be strategically applied to provide a pathway toward spatially complete and temporally continuous global flood monitoring?



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## 2. Overview of Satellite-Based Sensors

The sensors selected to be featured in this review are all currently operating in orbit. Importantly, their data are not restricted and can be freely accessed, encouraging widespread use and research that is open, reproducible, and replicable. We closely examine the following sensor types:

1. Multispectral sensors. This section encompasses classic spectral index-based water body mapping and thermal anomaly mapping from both Low Earth Orbit (LEO) and Geostationary Earth Orbit (GEO) satellites.
2. Microwave sensors. This section delves into both active radar and passive radiometry, including Global Navigation Satellite System Reflectometry (GNSS-R).
3. Altimetric sensors. This section focuses on altimetry over inland water bodies, which varies from open ocean altimetry.

Table 1 lists the satellite missions and sensors discussed in this review, in addition to their relevant characteristics. Sensor acronyms and other abbreviations in the text are defined in Table S1. Section 2 explores each of the sensor types and their capabilities for inundation monitoring, in addition to their limitations. Section 3 discusses the distinction between operational and scientific missions, followed by a quantitative assessment of each sensor's potential to capture flood events using the DFO flood database. Finally, Section 4 highlights how integrating data from various missions can overcome individual sensor limitations, and discusses potential directions for future research in satellite-based flood monitoring.

### 2.1. Multispectral

Multispectral remote sensing data are popular for water surface monitoring as they are freely available from multiple sources, often with reasonable temporal and spatial frequencies. The information held in the visible and infrared channels, commonly referred to as the optical spectrum, is sufficient for water body segmentation. Certain multispectral satellites have the capability to measure the response in the thermal portions of the spectrum, allowing them to keep track of important environmental metrics, such as land surface temperature. While it is not used as frequently as optical imagery, inundation monitoring with thermal imagery has been investigated (e.g., Burke et al., 2016; Leblanc et al., 2011). While passive microwave instruments also operate across multiple spectral bands, they are explored in the following section. The primary focus of this section concerns inundation monitoring with the reflective (visible, near-infrared, short-wave infrared) and thermal (mid-wave infrared, thermal infrared) portions of the electromagnetic spectrum. Hyperspectral imagery, though highly detailed, is primarily used for water quality analysis rather than extent mapping and will not be explored in depth in this review (see Arias et al., 2025; Bresciani et al., 2022; Cao et al., 2022 for more information). The following sections will further explore the application of multispectral sensors to water surface monitoring, exploring both optical and thermal sensing in this context.

#### 2.1.1. Optical

Generally, surface water extraction with optical satellite data is based on the spectral characteristics of water and other land cover types in the visible (VIS), near-infrared (NIR) and short-wave infrared (SWIR) channels (Figure 1). Water absorbs incident radiant flux in the SWIR and NIR channel and has a higher reflectance in the VIS channels. In contrast, vegetation strongly reflects NIR while bare soil reflects strongly in the SWIR channel, where the reflectance of water is near zero. These contrasting features permit the delineation of water bodies through exploitation of spectral properties in the SWIR and NIR channels.

Early on, Robinove (1978) demonstrated the usability of these spectral properties with an analysis of flooding in Australia utilizing the NIR channel of the Landsat-1 Multispectral Scanner, a LEO satellite. Surface water studies using the Landsat series also have the benefit of a high spatial resolution that enables detailed analyses from the last 40 years, beginning with 80 m in the Landsat-1 mission and progressing to 30 m in the recent missions equipped with the Operational Land Imager (Table 1). The Landsat series continues to be relevant today, allowing to generate historical records that are invaluable for understanding multi-decadal surface water dynamics, assessing flood risk, and informing water resource management strategies (Loomis & Koontz, 2024; Mueller et al., 2016; Pickens et al., 2020).

Another significant sensor for inundation monitoring from optical imagery is the Sentinel-2 MSI. Launched in 2015, the two-satellite Sentinel-2 constellation equipped with the Multispectral Instrument (MSI) captures

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**Table 1**
*All Sensors With Free and Publicly Available Data Available at the Time of Writing*

| Sensor type | Satellite | Sensor(s) | Revisit time | Spatial resolution | Geographical coverage | Launch year | Mission type | Landing page |
|---|---|---|---|---|---|---|---|---|
| Multispectral LEO | Landsat Legacy series | MSS<br>TM<br>OLI<br>TIRS | 8 days | 30–60 m | 82.6S to 82.6N | 1972 | Scientific | https://science.nasa.gov/mission/landsat/ |
| | MetOp | AVHRR | Twice daily | 1.1 km | Global | 1981 | Operational | https://navigator.eumetsat.int/product/EO:EUM:DAT:METOP:AVHRRL1 |
| | Aqua and Terra missions | MODIS | Twice daily | 250 m–1 km | Global | 1999 | Scientific | https://modis.gsfc.nasa.gov/ |
| | Suomi NPP | VIIRS | Twice daily | 375 m | Global | 2011 | Operational | https://ncc.nesdis.noaa.gov/VIIRS/ |
| | Sentinel-2 | MSI | 5 days | 10 m | 56S to 82.8N | 2015 | Scientific | https://sentinels.copernicus.eu/copernicus/sentinel-2 |
| | Sentinel-3 | SLSTR<br>OLCI | 0.5–3 days | 300–1 km | Global | 2016 | Operational | https://sentinels.copernicus.eu/copernicus/sentinel-3 |
| Multispectral GEO | Meteosat series | SEVIRI<br>FCI | 5–15 min | 0.5–5 km | 80N to 80S and 80W to 80E | 2002 | Operational | https://www.eumetsat.int/our-satellites/meteosat-series |
| | Himawari series | AHI | 5–15 min | 0.5–1 km | 60S to 60N and 180E to 160W | 2015 | Operational | https://www.jma.go.jp/jma/jma-eng/satellite/himawari89.html |
| | GOES-R series | ABI | 5–15 min | 0.5–1 km | 75S to 75N and 150E to 5W | 2016 | Operational | https://www.goes-r.gov/ |
| Passive Microwave Radiometer | DMSP series | SSM/I<br>SSMIS | Twice Daily | 25.0 × 12.5 km (19 GHz) | Global | 1987 | Operational | https://www.ncei.noaa.gov/products/satellite/defense-meteorological-satellite-program |
| | SMOS | MIRAS L-band radiometer | 1–3 days | 25 × 25 km (1.4 GHz) | Global | 2009 | Scientific | https://earth.esa.int/eogateway/missions/smos |
| | GCOM-W | AMSR2 | Twice Daily | 7 × 12 km (36.5 GHz) | Global | 2012 | Operational | https://global.jaxa.jp/projects/sat/gcom_w/ |
| | SMAP | L-band radiometer | 2–3 days | 36 km (1.4 GHz) | Global | 2015 | Scientific | https://smap.jpl.nasa.gov/ |
| Active Microwave Radar | Sentinel-1 | C-band SAR | 6 days | 10 m | 80S to 80N | 2014 | Operational | https://sentinels.copernicus.eu/copernicus/sentinel-1 |
| GNSS-R | CYGNSS | DDMI | 8 days | ~25 km × 25 km (incoherent), ~0.6 km × 6.6 km (coherent, theoretical minimum) | 38S to 38N | 2016 | Scientific | https://podaac.jpl.nasa.gov/CYGNSS |



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**Table 1**
*Continued*

| Sensor type | Satellite | Sensor(s) | Revisit time | Spatial resolution | Geographical coverage | Launch year | Mission type | Landing page |
|---|---|---|---|---|---|---|---|---|
| Laser Altimeter | ICESat series | GLAS<br>ATLAS | 91 days | 17 m swath | 82S to 82N | 2003 | Scientific | https://icesat-2.gsfc.nasa.gov/mission |
| Radar Altimeter | Jason series | Poseidon-3b Dual Ku-/C-band radar altimeter | 10 days | 5.2 km swath | 66S to 66N | 2001 | Operational | https://www.eumetsat.int/our-satellites/jason-series |
| | Sentinel-3 | Ku-/C-band SRAL | 27 days | 330 m | Global | 2016 | Scientific | https://sentinels.copernicus.eu/copernicus/sentinel-3 |
| | Sentinel-6 | Poseidon-4 Dual Ku-/C -band radar altimeter | 10 days | 11 km swath | 66S to 66N | 2020 | Operational | https://sentinels.copernicus.eu/missions/sentinel-6 |
| | SWOT | KaRIn | 21 days | 50 km swath with pixel sizes ranging 10–60 m | 78S to 78N | 2022 | Scientific | https://swot.jpl.nasa.gov/ |

*Note*. The launch year is associated with the launch of the first in the series, whereas the indicated sensors are currently in operation. For altimetric sensors, the values represent swath width rather than pixel resolution, as these instruments measure along a track rather than producing gridded imagery; where applicable, pixel sizes are noted separately (e.g., SWOT). The spatial resolution varies with the frequency for Passive Microwave Radiometers; the spatial resolution for the frequencies most referred to in the literature for flood applications are highlighted.



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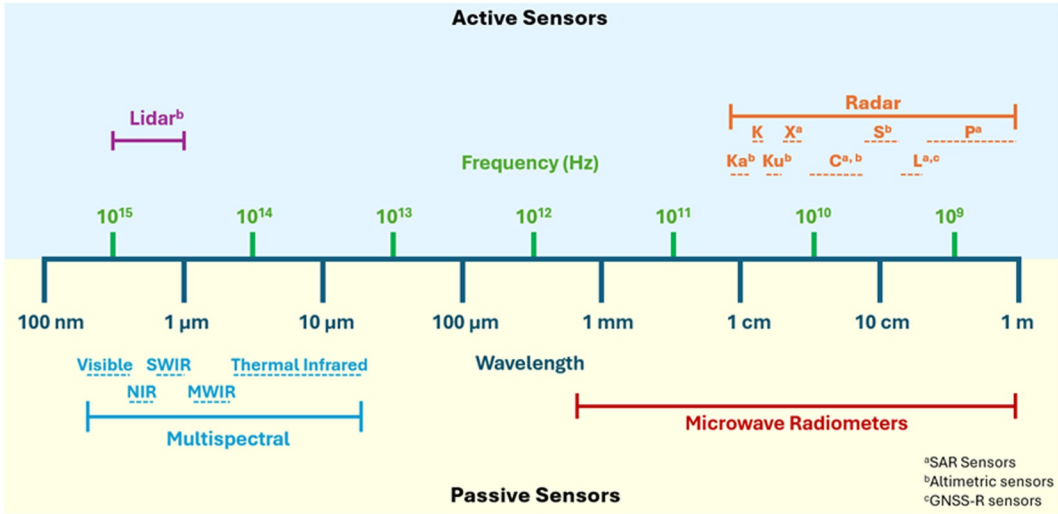



**Figure 1.** Frequency and wavelengths of sensors. NIR = near-infrared, SWIR = short-wave infrared, MWIR = mid-wave infrared. The specific wavelengths utilized for SAR, altimetry, and GNSS-R sensors are indicated. Figure adapted from Meng et al. (2024).

imagery globally with a spatial resolution of up to 10 m and a temporal resolution of 5 days, nearly a third of Landsat's (Table 1). Sentinel-2 MSI is one of the preferred sensors for surface water studies with the advantageous combination of a high spatial and moderate temporal resolutions, allowing it to frequently capture surface water dynamics. Similar to Landsat, the high spatial resolution enables the use of spectral indices that extract water bodies on a pixel scale basis, such as the NDWI and the Automated Water Extraction Index (Feyisa et al., 2014; Gao et al., 2012; McFeeters, 1996).

However, imagery every 5 days is not always sufficient, especially with the presence of cloud cover during acquisitions, such as in the tropical belt where floods are generated by seasonal monsoons. This limitation pushes some to utilize satellite sensors that operate at a higher temporal frequency, at the expense of a lower spatial resolution. Pixel-level water segmentation is not reasonable at very coarse spatial resolutions; thus, water monitoring occurs with sub-pixel-based methods. For coarser-resolution optical images, spectral mixture analyses are common solutions to extract meaningful information about the quantity of surface water in a large pixel. Sheng and Xiao (1994) facilitated water body monitoring through a mixed-pixel analysis by defining a ratio between AVHRR channels to obtain the ratio of water to land in each 1-km pixel at a sub-daily frequency. This principle was expanded upon and applied to data from MODIS to monitor daily water extent changes within a 500-m pixel. If the water body being monitored is relatively large, it may also be evaluated on a pixel rather than sub-pixel scale basis with the 250 m VIS and NIR bands. Users of MODIS imagery benefit from a daily repeat, making it a versatile and useful option for flood monitoring (Brakenridge & Anderson, 2006). Additionally, MODIS has been in orbit for over two decades, allowing long-term studies of hydrologic regimes (Petrakis et al., 2022). Another relevant sensor in water surface monitoring is VIIRS, which was launched in 2011 to support data continuity from aging AVHRR and MODIS. At a 375-m spatial resolution, it provides water coverage information at a higher level of detail. These LEO-based optical sensors are very commonly used for water monitoring, however, there are some non-traditional optical sensors that can provide valuable hydrological information. For example, Wang et al. (2019) demonstrated the utility of the Sentinel-3 OLCI to provide daily sub-pixel water body maps.

Sensors aboard GEO satellites are typically used for Near Real Time (NRT) weather monitoring, offering high temporal resolution and low data latency, making them suitable for tracking highly dynamic surface water phenomena. Proud et al. (2011) proposed an alternative to MODIS-based flood mapping with the MSG SEVIRI. By applying the Bidirectional Reflectance Distribution Function to track variations in reflectance, this method facilitates rapid tracking of large-scale flood events. Using optical data from the GOES-R series in combination with VIIRS enabled a clear-sky observation of the flooding from Hurricane Harvey in 2017 (Goldberg et al., 2018). GOES's high temporal resolution of 5 min effectively overcame cloud cover limitations to provide



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continuous flood maps during the event. A significant major advantage of GEO optical data lies in its potential to capture flood peaks, despite their coarse spatial resolution. The frequent acquisitions increase the likelihood of capturing peak flood levels, which is normally an unrealistic expectation when mapping floods from satellite-based sensors. The recently launched MTG series will provide GEO data over Europe and Africa, the GOES-R series covers the Americas, and the Himawari series provides imagery over Asia and Australia, freely available through the NOAA Open Data Dissemination Program (Table 1).

#### 2.1.2. Thermal

Another approach for water surface monitoring with multispectral sensors is based on the thermal end of the spectrum, particularly using the thermal infra-red channel (TIR). While utilized far less than optical imagery, thermal imagery has the potential to illustrate spatiotemporal changes in the water surface. Water's high thermal inertia and emissivity result in relatively uniform surface temperatures. The temperature stability, in contrast to the diurnal thermal fluctuations of dry land, allows for discrimination and mapping of water bodies based on thermal differences. This has been exploited successfully using data from both GEO and LEO multispectral satellites; however, the literature primarily utilizes temperature data with a coarse spatial resolution. Leblanc et al. (2011) successfully mapped Lake Chad's inundation patterns using maximum thermal composite data processed from MSG-SEVIRI, though the method was less effective during the rainy season due to similar temperatures between wet soils and flooded areas. Similarly, Burke et al. (2016) and Ordoyne and Friedl (2008) used a similar approach with the MODIS Land Surface Temperature (LST) and Emissivity product to evaluate inundation patterns in wetland landscapes. Finally, Li et al. (2023) employed the Sentinel-3 SLSTR in their multi-sensor flood mapping framework.

#### 2.1.3. Limitations

Imagery from multispectral sensors offers versatile and valuable data, providing a significant historical record for analyses. However, several obstacles, summarized in Table 2, can hinder its use. A primary challenge is cloud cover, which significantly reflects sunlight in the visible, near-infrared, and thermal infrared channels, effectively obscuring land surface information in both optical and thermal imagery. This issue complicates the use of optical imagery-based water body data sets, creating an inherent bias toward cloud-free conditions, often associated with dry periods. Consequently, crucial changes in surface water extent during the rainy season may be missed. Mercury et al. (2012), Leinenkugel et al. (2013) and later Pekel et al. (2016) examined the prevalence of cloud cover in optical imagery and found that tropical regions stand out as particularly cloudy, especially Southeast Asia, which suffers from an average of 85%–95% cloud cover during the rainy season. Globally, this is a marked issue for extreme inundation events, as these generally coincide with prolonged or heavy precipitation and associated cloudy conditions. These factors restrict the ability of multispectral sensors to fully capture the dynamics of flood events, especially rapid-onset events like flash floods, and make it difficult to pinpoint the precise timing of flood peaks.

Another difficulty lies in the detection of water under dense vegetation canopies. This issue is prevalent in many river basins such as the Amazon and Congo river basins, where significant portions of the wetland vegetation consist of flooded forests (Mayaux et al., 2002; Mertes et al., 1995). However, thermal-based methods have shown promise in overcoming this limitation, allowing for observation of inundation patterns in wetland regions, albeit at a coarse spatial resolution (Leblanc et al., 2011; Ordoyne & Friedl, 2008). More recently, researchers began incorporating auxiliary topographic information that can help fill the gaps imposed by dense vegetation cover (Jiang et al., 2021; Li & Demir, 2024).

Finally, varying spatial and temporal resolutions of the various sensors can limit their applicability. Sensors with higher spatial resolutions, such as those on the Landsat series and Sentinel-2, can deliver highly detailed hydrologic information, provided that the cloud and vegetation cover is not too extensive. However, their infrequent data acquisition can miss significant processes, such as flooding. Consequently, a tradeoff exists between high spatial, low temporal resolution sensors and low spatial, high temporal resolution sensors. These differing spatiotemporal scales dictate the precision and utility of flood detection for specific applications. High spatial resolution (<30 m) data are most appropriate for local scales and is particularly useful for insurance and land use planning related applications that do not require low-latency, rapid data (Blyth, 1997; De Leeuw et al., 2014).

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**Table 2**
*Primary Strengths and Limitations of Each Sensor Type*

| Sensor type | Strengths | Limitations |
|---|---|---|
| Multispectral LEO | • Historical record | • Cloud cover |
| | | • Vegetation obscuration |
| | • Medium to high spatial resolution (20–500 m) | • Low temporal resolution (5–16-day repeat) |
| Multispectral GEO | • Historical record | • Cloud cover |
| | | • Vegetation obscuration |
| | • High temporal resolution (5–15 min) | • Low spatial resolution (500–2 km) |
| Passive Microwave Radiometry | • High temporal resolution | • Low spatial resolution (10–50 km) |
| | • Not impacted by clouds | |
| | • Historical record | • May overestimate surface water fraction in some conditions, particularly after heavy rain events when wet soils increase brightness temperature polarization differences (e.g., Galantowicz & Picton, 2021) |
| Active Microwave Radar | • High spatial resolution (10–20 m) | • Low temporal resolution (6–12 days) |
| | | • Double-bounce effects |
| | • Not impacted by clouds | • Sensitivity to high winds |
| | | • Speckle (image noise) |
| GNSS-R | • Medium to high temporal resolution (sub-daily to weekly) | • Limited historical record |
| | | • Irregular and coarse spatial sampling (0.6–25 km depending on geometry) |
| | • Not impacted by clouds | • Water fraction overestimates in areas with wet soils |
| Laser Altimetry | • Extremely high precision | • Low temporal resolution (>90 days) |
| | • Applicable to small water bodies | • Limited swath coverage |
| Radar Altimetry | • Not impacted by clouds or adverse weather conditions | • Low temporal resolution (10–35-day repeat) |
| | | • Off-nadir pulses observe the surrounding terrain rather than the water itself |
| | | • Only works well for large water bodies, apart from SWOT |

Conversely, medium (30–500 m) and low (500 m–10s of kilometers) spatial resolution sensors find their utility linked to emergency response and large-scale floodplain mapping (Blyth, 1997; Schumann, 2021).

### 2.2. Microwave

Sensors in the microwave portion of the electromagnetic spectrum are versatile sensors that can capture information in clouded weather conditions, day, or night, making them the sensor of choice in many applications. Microwave sensors may be active or passive, each with distinct characteristics and abilities for recording earth surface processes. Microwave radar sensors measure long wavelengths in the microwave portion of the spectrum, increasing their penetration of the signal into objects such as clouds, snow, and vegetation. The following section expands upon active microwave sensors, in the form of SAR or GNSS-R, and passive microwave (PMW) sensors.

#### 2.2.1. Synthetic Aperture Radar

Synthetic Aperture Radar (SAR) sensors are active microwave sensors that can penetrate cloud cover and operate day and night, providing data under nearly all weather and light conditions. Unlike Real Aperture Radar (RAR), whose antenna bandwidth scales with the target distance, SAR uses signal processing to synthesize a larger, virtual antenna. This enables high-resolution satellite-based imaging that is not feasible with RAR. SAR operates under the radar principles, where the range to the target is derived using the time delay of the backscattered microwave pulse (Long & Ulaby, 2015). The return signal to the sensor, the echo or backscatter signal, is a function of the surface roughness, incidence angle, size of scatterers in the target area, dielectric constant, wavelength, and polarization of the SAR pulses (Meyer, 2019). These parameters contain valuable information about the physical characteristics of the Earth's surface and are particularly sensitive to changes in hydrologic

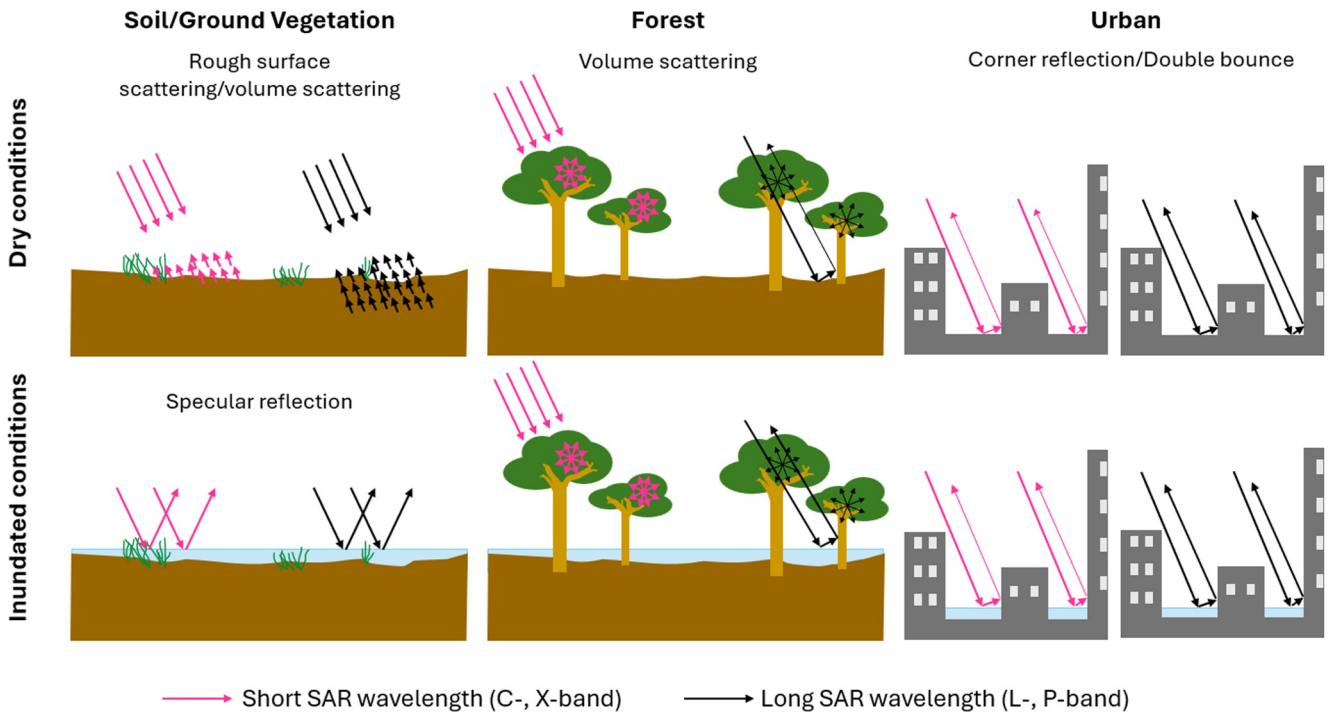



**Figure 2.** Simplified schematic illustrating the various scattering mechanisms under dry and wet conditions for both short (pink) and long (black) SAR wavelengths.

conditions. The scattering mechanism, influenced by target object properties, can be categorized as surface, volume, or hard-target scattering. Radar pulses scatter in several ways when encountering a surface, influenced by both the surface properties and the medium they traverse, as depicted in Figure 2.

In situations with bare soil and minimal ground vegetation, the backscatter signal varies significantly depending on the presence of water. When dry, there is surface reflection at both short and long wavelengths. Short wavelengths experience some degree of volume backscatter from the grasses while longer wavelengths penetrate more deeply into the soil (Schumann & Moller, 2015). In cases where water is present and there is minimal surface roughness, both SAR wavelengths exhibit specular reflection. Specular reflection, akin to a mirror, is particularly useful for detecting water bodies. The smooth surface and high dielectric constant of water cause the radar beam to reflect specularly, like a mirror, resulting in a low backscatter signal. Vegetation exposed above the water can additionally contribute to backscattering at shorter wavelengths (Schumann & Moller, 2015). In the second scenario of Figure 2, short SAR wavelength volume scattering is dominant within the tree canopies, and the radar pulse may not be able to pass through to the surface. Longer wavelengths may pass through to the water surface under inundated conditions, creating a double-bounce effect between the water surface and tree trunk. Finally, in urban settings, the double-bounce effect is dominant at both wavelengths and is particularly strong under flooded conditions.

Beyond single SAR images, multiple SAR images may be analyzed in combination using Interferometric Synthetic Aperture Radar (InSAR) to detect and monitor surface water. Frequently used in topography-related studies, InSAR relies on the phase change between two or more SAR image acquisitions to isolate surface deformations. Coherence, the normalized correlation between the SAR images, provides information about the temporal stability of scattering mechanisms on the ground surface (Refice et al., 2014). These data can be used as an additional layer of information to help differentiate between flooded and non-flooded areas. Water detection with InSAR is frequently applied for wetland monitoring, as it can determine seasonal inundation variations under dense wetland vegetation (Liao et al., 2020). It can be helpful for flood mapping as it quickly isolates significant changes to the surface terrain, such as an influx of floodwater. This method can enhance floodwater mapping in urban areas, helping to overcome multiple-bounce scattering issues, by exploiting changes in both Vertical Transmit-Vertical Receive (VV) and Vertical Transmit-Horizontal Receive (VH) polarizations (Chini et al., 2019; Pelich et al., 2022). Additionally, InSAR can provide information regarding water level changes with



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Differential InSAR (D-InSAR). Regarding inland water bodies, D-InSAR computes the phase difference between a pair of SAR images to yield water level variations. This approach is particularly effective in wetland environments with dense vegetation, small lakes, and shorelines. The backscatter from vegetation increases coherence, capturing relative phase variations between pixels and yielding centimeter-scale vertical displacement measurements (Aminjafari et al., 2024).

Historically, SAR's potential for water monitoring was initially realized with NASA's Seasat mission, launched in 1978, and equipped with an L-band SAR sensor. Early innovations with Seasat such as from Macdonald et al. (1981), demonstrating the sensitivity of Seasat's L-band radar to changes in water levels beneath forest canopies, and Foster and Hall (1981), showing its suitability for detecting hydrological drainage features, enabled SAR to quickly become a valuable resource for hydrologic information. Today, SAR is an essential component of hydrological studies with Sentinel-1 as the primary open-source and publicly accessible SAR satellite to provide water extents. Designed as a twin satellite constellation, Sentinel-1B's system failure in 2022 reduced the revisit time of Sentinel-1 to 12 days. However, the recent launch of Sentinel-1C in December 2024 brought this time down to 6 days again. In the interim period between Sentinel-1B's failure and Sentinel-1C's launch, the remote sensing community has relied primarily on Sentinel-1A. Despite this, there are still significant developments in SAR-based water monitoring in recent years based on Sentinel-1. Notable improvements in water detection and mapping in arid regions, where both sand and water surfaces exhibit low backscatter, with the incorporation of SAR textures or by fusing VV coherence and amplitude information (Garg et al., 2024; Ritushree et al., 2023). Other SAR sensors currently in orbit, such as DLR's TerraSAR-X, Italy's COSMO-SkyMed, and the Canadian Space Agency's Radarsat-2, have data access restrictions due to the dual-use nature of the missions. Imagery may be provided freely upon request for scientific use but global free release for automated monitoring is hardly possible, thus these missions are not suitable data sources for global users.

#### *2.2.1.1. Limitations*

Users of SAR should be cognizant of certain obstacles when extracting water-related information. SAR imagery has geometric and radiometric properties that require substantial preprocessing and are prone to error. Imaging radar systems operate at an oblique angle, leading to geometric distortions such as foreshortening, layover, and shadows (Meyer, 2019). These distortions, especially shadows, can be very difficult to rectify in mountainous topography. When using SAR imagery for water surface extent purposes, flooded vegetation and buildings may create double bounce effects and volume scattering within vegetation, contributing to increased backscatter levels and consequently water classification errors (Landuyt et al., 2019). Wet soils and smooth asphalt surfaces also exhibit similar specular reflection characteristics to water, leading to some misclassifications (Munasinghe et al., 2023). Furthermore, cities may experience complications with significant layover adjacent to buildings, causing some streets to be misclassified as dry when they are actually inundated (Mason et al., 2012). Additional challenges are present in SAR imagery acquired under significantly windy conditions, as this will cause the water surface to roughen up and give a heightened backscatter response that is more similar to dry areas (Landuyt et al., 2019). These known limitations aid generation of vast exclusion masks, where SAR data are not reliable (Misra et al., 2025; Seewald et al., 2023). InSAR's potential is maximized when the time between SAR image acquisitions is minimized. Using imagery acquired in short time intervals reduces atmospheric effects and temporal decorrelation, and increases the chances of accurately capturing rapid ground deformation events, such as floods (McCormack et al., 2011). Current public SAR satellites have relatively low temporal resolutions, restricting InSAR's capability to monitor and capture rapid deformation.

### 2.2.2. Global Navigation Satellite System Reflectometry

Global Navigation Satellite System Reflectometry (GNSS-R) is typically used for evaluating soil moisture, vegetation monitoring, altimetric measurements, and wind speed estimates over oceans; however its application to surface water monitoring is a more recent development (Ferrazzoli et al., 2011; Rius et al., 2012; Rodriguez-Alvarez et al., 2011, 2013). The detection and monitoring of surface water bodies with GNSS-R surface-reflected signals is based on active L-band bistatic radar data recorded by the GNSS-R receivers in the form of a Delay-Doppler Map (DDM). DDM values represent the correlation between the surface-reflected signal and a replica signal stored within the receiver. At L-band frequencies, the DDM-derived reflectivity is quantitatively determined by the soil's Fresnel reflectivity, which is a function of its wetness-dependent dielectric constant that is in turn attenuated by the surface roughness (Dobson et al., 1985; Roo & Ulaby, 1994). Building upon this

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knowledge, Chew et al. (2018) quantified the relationship between the L-band received surface reflectivity and the surface water extent. Substantial increases in surface reflectivity are indicative of larger areas of ponded water, enabling the mapping of water bodies.

NASA's CYGNSS constellation of microsatellites, launched in late 2016 to monitor ocean near-surface wind speeds in the tropical cyclone-prone regions, has proven to be monumental in the application of GNSS-R to water monitoring applications. As an active radar system, CYGNSS is capable of detecting surface water under all weather conditions and can penetrate above-ground vegetation to some extent (Yueh et al., 2022). Since this constellation was not designed for land surface monitoring, the revisit time over land is irregular and approximately every 3–8 days, significantly longer than its roughly 7-hr revisit time over oceans (Ruf et al., 2016). CYGNSS footprints are large, measuring a minimum of 3.5 × 0.5 km, and exhibit irregular spacing. Despite its large footprint, CYGNSS is capable of detecting water feature changes on scales as small as 100–200 m wide due to the coherency of the reflected signal over smooth surfaces, rendering it an effective technique for surface water monitoring (Chew et al., 2023). To address the gaps in spatial coverage from the irregular revisit time, the surface reflectivity may be interpolated onto a gridded map. This gridded data, combined with soil moisture and surface roughness information for both land and water, enables derivation of a water fraction map indicating the percent of the 3 km × 3 km pixel that is covered by surface water on nearly a daily basis (Chew, 2023). Pu et al. (2024) took a different approach, using computer vision to generate monthly inundation maps at a 0.01° spatial resolution over the entire CYGNSS domain. An analysis by Downs et al. (2023) found that GNSS-R derived inundation maps were comparable to Sentinel-1 SAR, offering a similar advantage over optical-derived maps.

#### *2.2.2.1. Limitations*

GNSS-R, particularly the CYGNSS constellation, is emerging as a valuable tool for inundation monitoring given its ability to frequently detect surface water changes with a smaller footprint than PMW data. However, its applications are currently limited by several factors. First, CYGNSS's orbit is confined to tropical cyclone impacted regions between 38°N and 38°S latitudes, restricting global coverage. Second, the irregular spacing and repeat time over land often requires researchers to perform spatial interpolation on the surface reflectivity data, which can introduce uncertainties, especially in areas with significant anomalies. Finally, the large footprint and sensitivity to water can lead to overestimates in the water fraction, particularly in areas with wet soils (Setti & Tabibi, 2024).

### 2.2.3. Passive Microwave Radiometry

Passive Microwave Radiometry (PMW) is a long-established technique for monitoring both atmospheric and Earth surface processes. PMW operates with long wavelengths in the microwave portion of the spectrum, collecting data under clouded conditions and darkness (Figure 1). PMW passively acquires brightness temperature (K) measurements, which are directly proportional to the product of effective land surface temperature and emissivity when the atmospheric contribution is limited. Primarily used to monitor properties such as soil moisture, sea surface temperature, sea ice concentration and atmospheric conditions, PMW can be used to observe changes in surface water extents at a sub-daily temporal frequency, at the expense of a very coarse spatial resolution, often greater than 20 km per pixel. PMW sensors are generally equipped with a 36.5–37 GHz channel that receives a large signal for inundation relative to other land surface processes. However, at this frequency, atmospheric water vapor and temperature can impact the brightness temperature measurements. To lessen these influences, the difference between the vertically and horizontally polarized brightness temperatures, denoted as $\Delta T$, is often applied for hydrological studies instead of the actual brightness temperatures (Choudhury, 1989; Smith, 1997). $\Delta T$ values are high over open water due to its strongly polarized microwave signal. However, this signal is weakened by factors that cause depolarization such as scattering from vegetation, surface roughness, and precipitating clouds, as well as interference from atmospheric gas like water vapor, will decrease $\Delta T$ values (Choudhury, 1989). These trends suggest that $\Delta T$ decreases with increasing vegetation density and surface roughness, while increasing over wet or inundated soils.

Choudhury (1991) demonstrated the potential of $\Delta T$ as a sensitive indicator for monitoring surface water changes to monitor seasonal surface water changes in South America using the SMMR. Sippel et al. (1994) expanded on the use of the SMMR at 37 GHz for hydrological monitoring by applying a mixing model to estimate the proportion of each pixel covered by surface water. The works of Galantowicz (2002) and De Groeve et al. (2007)

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further applied the water surface pixel fraction to flood monitoring applications while Brakenridge et al. (2005) derived a methodology for river discharge estimation using the AMSR-E, linking a larger water surface area to a higher stage and thus increased river discharge. AMSR-E's successor, AMSR2, continues to provide valuable PMW data for environmental monitoring (Du et al., 2017; Mizuochi et al., 2014). The SMMR's following sensors, the Special Sensor Microwave/Imager (SSM/I) and subsequently the Special Sensor Microwave Imager/Sounder (SSMIS) have provided a lengthy historical record of hydrological observations for various applications (Prigent et al., 2007; Schroeder et al., 2015; Takbiri et al., 2017). Further advances have been made with the SMAP mission, where the L-band radiometer observations can be used to derive both discharge (Kugler et al., 2024) and water fraction estimates (Du et al., 2018). The SMOS mission's L-band has also provided valuable inundation information in densely vegetated regions, such as Southeast Asia (Tangdamrongsub et al., 2021) and the Amazon River Basin (Parrens et al., 2017). In addition to these innovations, further work used topographic information to disaggregate the water fraction within a pixel, resulting in higher resolution maps of water bodies (Fluet-Chouinard et al., 2015; Galantowicz & Picton, 2014; Nguyen & Aires, 2023; Zheng et al., 2017).

#### *2.2.3.1. Limitations*

PMW-based water monitoring is primarily limited by its coarse spatial resolution. Topography-based downscaling can enhance the spatial resolution, but the accuracy of the water distribution estimates is constrained by the inherent errors in global DEMs, particularly in regions with complex topography (Mukherjee et al., 2013). Notable obstacles are present in coastal environments, where complex geomorphologic features such as estuaries, tidal flats, and river deltas may not be clearly identified with coarse imagery. Topographic model quality in coastal areas is generally poor, limiting the application of water fraction downscaling methods (Emmendorfer et al., 2024). Additionally, the large scale at which PMW senses limits its ability to detect and monitor small-scale hydrology. Finally, there is the possibility of significant and unrealistic jumps in the PMW-derived water fraction during and after large rainstorms, likely related to the effects of soil moisture (Galantowicz & Picton, 2021).

### 2.3. Altimetry

Satellite-based altimetry is a remote-sensing technique that provides water surface elevation measurements using the two-way travel time of radar or laser pulses and the signal of the reflected water surface. Altimetry measurements may be made with a radar altimeter, which emits microwave radar pulses, or with a laser altimeter, which instead emits light pulses.

#### 2.3.1. Radar Altimetry

Radar altimetry operates by emitting a radar pulse in the microwave range to the surface and calculating the time it takes for the reflected backscatter to return to the sensor, measuring the height of surface water. Radar altimetry over inland water bodies differs from radar altimetry over oceans. Ocean surfaces are typically rough, rippled by waves and currents, and composed of numerous small, angled facets. Each facet reflects a portion of the radar pulse, and the altimeter receives the sum of these reflections. In contrast, inland water bodies generally possess smoother surfaces that specularly reflect the radar pulse. This characteristic enables precise water height measurements for large lakes and rivers, given the large 80–300 km inter-track distance that altimeters possess. Radar altimeters typically operate at Ka-, Ku-, C-, and S-band frequencies (Figure 1). Current altimeters, such as from the Jason series, Sentinel-3, and Sentinel-6, perform altimetry measurements with a dual frequency Ku- and C- band radar altimeter. Dual frequency altimeters correct the ionospheric path delay in the Ku-band range estimate using the collocated ranging measurement (Papa et al., 2003).

While radar altimetry was originally designed for ocean and cryosphere studies, it has been successfully applied to inland water for decades, enabling remote global monitoring of rivers and lakes. Early surveys of inland water bodies utilized TOPEX/POSEIDON data and demonstrated the successful application of radar altimetry to provide information where gauge data are not available or lakes are inaccessible for regular monitoring (Birkett, 1995). Currently, radar altimetry is used to determine inland water levels, volume variations, and river discharge (Crétaux et al., 2015, 2018; Gao et al., 2012; Tarpanelli et al., 2013). It is particularly useful for river discharge estimations in poorly gauged regions to aid hydrologic model refinement (Getirana et al., 2009). As a result, altimetry plays a significant role in flood forecasting systems (Falck et al., 2021).

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The recently launched SWOT mission innovatively uses a Ka-band radar imaging altimeter with a footprint 1,000 times smaller than that of conventional altimeters, providing surface water levels and corresponding water storage changes with a spatial resolution of approximately 100 m (Fu et al., 2023; Papa & Frappart, 2021). The SWOT science team uniquely develops a "pixel cloud" water mask product. Unlike traditional radar altimeters that deliver isolated height measurements at specific points, this innovative product provides a comprehensive rasterized water mask with associated geolocated heights, an unusually comprehensive observation. However, while it has promising flood monitoring capabilities, its limited temporal sampling and the high latency in releasing some products restrict its usefulness in this capacity. Frasson et al. (2019) estimated that SWOT would capture 55% of the flood events recorded by the Dartmouth Flood Observatory. This finding is consistent with Nickles et al. (2019) and Cerbelaud et al. (2024), who concluded that capturing peak flood discharge with SWOT would be rare, relying primarily on opportunistic observations. However, when SWOT data are available, it tracks discharge variations with accuracy (Andreadis et al., 2025).

#### 2.3.2. Laser Altimetry

Laser altimetry, also known as LiDAR, is primarily used to generate highly detailed and accurate topographic data sets. LiDAR measures surface elevations by emitting laser pulses and calculating the time it takes for the reflected light to return to the sensor. Regarding the application of LiDAR to water surface monitoring, the most success has been found with the ICESat missions. The ICESat missions are exceptionally precise with a high vertical accuracy of up to 5 cm and have a smaller footprint than that of radar altimetry missions, enabling measurements of small (<1 $km^2$) water bodies (Brunt et al., 2019). ICESat-1's high measurement precision has also been applied to measure river water surface slopes and correct river discharge gauges (Hall et al., 2012; O'Loughlin et al., 2013). Many researchers have taken advantage of ICESat-2's small footprint to monitor water surface levels on lakes and reservoirs globally (Chen et al., 2022; Lao et al., 2022; Xu et al., 2021). Despite its sporadic sampling, it has also been evaluated for flood applications when combined with other data, like that of Sentinel-2 (Y. Cao et al., 2023). Furthermore, laser altimetry is valuable for precise water surface elevation retrievals and hydraulic model calibration in the absence of in situ data (Musaeus et al., 2024; Zhou et al., 2023).

##### *2.3.2.1. Limitations*

Both radar and laser altimeters are constrained by their low temporal resolution, limiting the volume of data available for analysis (see Table 1). Regarding radar altimetry specifically, complex topographies and small water bodies can complicate measurements as its wider radar pulses may be reflected off-nadir, leading to observations of the surrounding terrain rather than the water itself (Maillard et al., 2015). Radar altimeters typically have a large footprint that is unable to detect and measure smaller water bodies, apart from the SWOT mission (Table 2).

## 3. Discussion

### 3.1. Challenges in the Remote Sensing of Inundation

The very nature of flooding presents a significant challenge to satellite-based monitoring. The fundamental issue in the remote sensing of inundation is the spatiotemporal mismatch between data acquisition and flooding itself. The ephemeral evolution of a flood is difficult to capture with discrete observations from satellites on fixed schedules.

Typically, extreme inundation events are associated with intense hydrometeorological phenomena, which come with cloud cover and sometimes high winds. These conditions inhibit the acquisition of useful information from most EO-based sensors. Even with optimal temporal alignment between sensor acquisition and a flood event, significant cloud cover can preclude the acquisition of usable data. Another complication is the insufficient temporal resolution of certain sensors. Although they can overcome atmospheric challenges by penetrating the atmosphere, they may fail to capture crucial details of damaging impacts, especially the flood peak. Flash floods generally last less than 6 hr; thus, the potential for cloud-free capture is rare (WMO, 2006). Other types of floods, such as those resulting from monsoon or river systems, tend to have longer durations. Globally, the median flood duration is estimated to be 8 days, with longer durations at the tropics and subtropics and shorter durations at higher latitudes (Najibi & Devineni, 2018; Tarpanelli et al., 2022). Longer flood duration increases the possibility of useful image capture; however, this is not guaranteed.

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### 3.2. Data Relevance for Hydrology

A primary difficulty in the remote sensing of inundation is the historical lack of hydrology-focused satellite missions. For decades, the field has advanced by repurposing data from missions designed primarily for meteorology, land cover mapping, or general water cycle studies. Parameters like terrestrial water storage or flood extent were often derivative afterthoughts rather than primary mission objectives.

The SWOT mission represents an important milestone as the first satellite designed and dedicated to observing terrestrial surface water. However, SWOT is a scientific mission that has a planned lifetime of 3.5 years and no guaranteed follow-on mission. The upcoming HydroGNSS constellation will be another example of a hydrology-focused mission, designed to use GNSS-R to evaluate hydrologically linked Essential Climate Variables (ECVs). Similarly, HydroGNSS is designed for a 3-year lifetime only.

Given the significant lack of terrestrial water-focused satellite-based data, researchers will need to continue using alternative and creative solutions using data from a variety of satellite missions with different intended applications. Given the number of limitations inherent to each type of sensor, their combination in a multi-sensor framework may help to overcome some of these limitations.

### 3.3. Sensor Operational Status

When designing a new methodology for water detection and monitoring from one or more sensors, a commonly overlooked element is the operational capacity of the satellite sensor mission. Typically, the operational status of a satellite refers to its functioning, or lack thereof, in orbit. In this case, however, we highlight the distinction between scientific missions and operational missions. Scientific satellite missions collect data primarily for research purposes, have no strict data latency requirements, and are not necessarily obligated to conduct follow-on missions to ensure continuity of the data.

Operational satellites have data continuity as a mission requirement; thus, they have guaranteed funding, reliability, and longevity (i.e., follow-on missions). Moreover, operational satellite missions can provide NRT data, which are essential for Numerical Weather Prediction (NWP). These missions, often managed by agencies with specific mandates, are obligated to report and disseminate data within strict time frames (Lahoz, 2010). Due to their role in NWP, operational satellites are generally under the purview of meteorological agencies, such as NOAA or EUMETSAT.

Additional considerations should be given to data access and processing. Operational missions typically have established distribution channels and highly standardized data formats, with the latency of the data clearly defined. On the other hand, scientific missions may not have clear data latency or distribution standards as a part of their mission parameters, limiting their applicability in disaster situations (Rolla et al., 2025). Taking Sentinel-2, for example: while its positioning products have a NRT delivery requirement, its imagery does not, making the data unreliable for time-sensitive flood response. In contrast, the high temporal frequency, low latency, and reliable data quality of the operational VIIRS, Himawari, and GOES-R satellite missions enabled Li et al. (2022) to map flooding in a truly operational sense, always providing flood insights without an emergency activation.

Both operational and scientific missions have important places in inundation applications, but users must consider their specific application needs, in particular, the longevity and latency that are required for their application. This distinction may be particularly important when evaluating sensors to utilize in extreme flood monitoring applications, as sensor imagery characteristics such as minimal latency, frequent updates, and maximal reliability are vital for proper disaster response coordination. Conversely, when very precise evaluations of specific inundation dynamics are required, a scientific sensor with a high spatial resolution and validated data may be the optimal choice.

### 3.4. A Quantitative Assessment of Sensor Efficacy Using the Global Flood Record

The Dartmouth Flood Observatory (DFO) flood events database contains a thorough list of global flood events from 1985 to late 2021 (*Dartmouth Flood Observatory*). Specifically, this flood event database identifies where flood events have occurred, records an estimated level of impact and the duration, and validates the flood occurrence, often with a news source. While remote sensing data are important components of the DFO, this particular database table does not include remotely sensed data. The flood events in the DFO database are, at a

**Table 3**
*Percent of DFO Database Flood Events With Viable Data From Each Sensor Since Each Sensor's Launch*

| Sensor | Launch year | Flood events with viable imagery available since launch (%) |
|---|---|---|
| ICESat series* | 2003 | 15 |
| Landsat Legacy series* | 1972 | 18 |
| Sentinel-2* | 2015 | 30 |
| Jason series◊ | 2002 | 37 |
| MODIS* | 1999 | 40 |
| VIIRS◊ | 2011 | 51 |
| SWOT* | 2022 | 55 |
| Sentinel-1◊ | 2014 | 61 |
| CYGNSS* | 2016 | 62 |
| Sentinel-6◊ | 2020 | 65 |
| Sentinel-3 OLCI◊ | 2016 | 70 |
| AVHRR◊ | 1981 | 71 |
| Himawari series◊ | 2015 | 81 |
| Meteosat series◊ | 2002 | 86 |
| GOES-R series◊ | 2016 | 86 |
| SMOS* | 2009 | 95 |
| SMAP* | 2015 | 97 |
| AMSR2◊ | 2012 | 97 |
| SSMI/SSMIS series◊ | 1987 | 98 |

*Note*. *Refers to sensors that are typically scientific, whereas ◊ refers to operational sensors.

minimum, large flood events with significant damage to structures or agriculture, collected using information from news, governmental, instrumental, and remote sensing sources. A uniquely comprehensive and global database, it is well-suited for analysis of the effectiveness of different satellite sensors in their capture of flood events. This database has known reporting, geographic, and threshold biases, and systematically underrepresents smaller or flash flood events. Therefore, these success rates reflect sensor performance for capturing these specific types of major, impactful floods. We estimated the potential data availability from each sensor for the locations and time periods identified in the DFO database, using the open-access platforms Google Earth Engine, NASA Earthdata, and data hosted by the NOAA Open Data Dissemination Program (Table 3). For each sensor, we searched for any usable data that coincided with the flood event location and duration. If there were data available, the flood event was marked as potentially observable for that particular sensor. This procedure indicates the potential of the various sensors to capture some flood signal and does not guarantee full, accurate detection and depiction of the flooding. It is quite possible that only a portion of the flood may be detected.

Following Munasinghe et al. (2023), we defined a usable optical image as one with 20% or less cloud cover. We acknowledge this threshold introduces a margin of error, as imagery with higher cloud cover may be partially useable, and our large-scale analysis may not automatically detect all sensor-specific quality errors. The sensors lacking global coverage (e.g., GOES-R) were only evaluated for events that were in their domain, as defined in Table 1. Furthermore, the value provided for SWOT was obtained from Frasson et al.'s (2019) work, seeing that the launch of SWOT occurred in 2022, after the DFO flood event database was no longer maintained. The results, recorded in Table 3, illustrate the percentage of flood events with potentially available data from each sensor, calculated starting from the launch of each sensor. High values indicate a high success rate for capturing data for the provided flood events, whereas low values indicate that the sensor historically struggles to capture events, whether due to temporal constraints and/or cloud cover.

The outcome of the assessment, summarized in Table 3, provides evidence that no single sensor alone is sufficient for reliable, detailed, large-scale flood monitoring. The results align with and validate long-held expert



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understanding. Previous studies suggested that SAR sensors like Sentinel-1 are roughly twice as effective as optical sensors like Sentinel-2 for flood detection in Europe due to cloud cover; the analysis of the DFO's real-world global event catalog confirms this relative performance (Misra et al., 2025; Tarpanelli et al., 2022). The high success rates of PMW sensors like AMSR2 and SMAP demonstrate their reliability for detecting large-scale hydrologic anomalies, but the lack of granularity is problematic for effective flood management. In contrast, the low success rates of high-resolution optical sensors like Landsat and Sentinel-2 highlight their vulnerability to cloud cover. This reveals a disconnect in the remote sensing of flood events as the sensors underrepresented in the literature for flood detection and mapping are in fact the most suitable to do so consistently. Utilization of these sensors in combination may then potentially increase our ability to evaluate inundation, despite the difficulties posed by the nature of flooding.

This assessment of data captured during global flood events, in combination with the knowledge of each sensors' limitations, cements that no single sensor or mission can adequately meet the requirements for comprehensive flood monitoring; however there is high potential for a flood signal to be captured by a sensor or variety of sensors. Temporal frequency constitutes the fundamental prerequisite for such synergy. Before sensors can complement each other physically, they must first observe the same event within a relevant timeframe. This reality necessitates a strategic approach that leverages the strengths of the various sensors, involving an understanding of the distinction between operational and scientific missions to ensure data reliability and latency are fit for purpose, and more importantly, embracing sensor synergy as the primary method for overcoming the observational gaps and insufficiencies inherent to any individual platform. The evolution of the remote sensing of inundation can be seen as progression from the opportunistic repurposing of data from single, non-hydrological missions to the intentional synthesis of data from multiple complementary systems.

### 3.5. Sensor Synergies

High-resolution single sensors alone can capture relevant information, but may still miss nearly a third of flood events (Munasinghe et al., 2023; Nigro et al., 2014). Our assessment of the DFO flood event database shows that Landsat and Sentinel-2 captured only 18% and 30% of major flood events, respectively, emphasizing a persistent gap due to revisit and cloud constraints. In contrast, PMW sensors such as the AMSR2 detected greater than 95% of flood events, but at the cost of local detail, validating the necessity of sensor fusion for achieving both coverage and precision. Data fusion and integration can potentially capture more useful information, relying on the complementary nature of different sensor data sources that fill each other's gaps.

As there are many types of sensors to choose from, it is valuable to objectively consider their strengths and weaknesses to ensure that the resulting data-fusion product meets requirements of an inundation-focused application. Figure 3 represents one way to visualize and compare the relative sensor strengths and limitations. The qualitative 1–5 scale was adopted to facilitate an intuitive comparison across sensor types with variable key metrics. Thresholds for each attribute are discretized as follows: temporal resolution ranges from <1 day (score 5) to >16 days (score 1); spatial resolution ranges from <10 m (score 5) to >10 km (score 1); atmospheric influence is scored from negligible (score 5) to very high/obscured (score 1); historical record from >30 years (score 5) to <3 years (score 1); and data latency from <1 hr (score 5) to >7 days (score 1). By comparing the attributes directly of the different sensors, it is possible to isolate where one sensor has a weakness and another has a strength, thus presenting a potential option for data fusion to fill information gaps. The vast number of satellite sensors now in orbit can capture flood events simultaneously, yielding to a wealth of information for each event (Figure 4). There is no lack of available data, rather, a possible lack of synergistic data-fusion flood monitoring systems.

Between 2015 and 2020, the median number of sensors collecting valid data for a single flood event effectively doubled (Figure 4). Notably, the Copernicus program satellites (e.g., Sentinel-1, Sentinel-2, and Sentinel-3) emerged in this period, enabling open-access high-resolution data. This was a particularly relevant breakthrough for SAR data, which was not previously so accessible (Schumann, 2024). Furthermore, the percentage of DFO events with observations reached 100% by 2005, indicating that every event in the DFO event database has observations available from at least one sensor from 2005 onwards. This illustrates that there is not a question of data availability, rather, a question of proper multi-source data usage to appropriately monitor and characterize flood events. A multi-sensor approach combining altimeter and optical data, for example, successfully characterized flood dynamics in the Bolivian Amazon by overcoming issues like dense vegetation and cloud cover

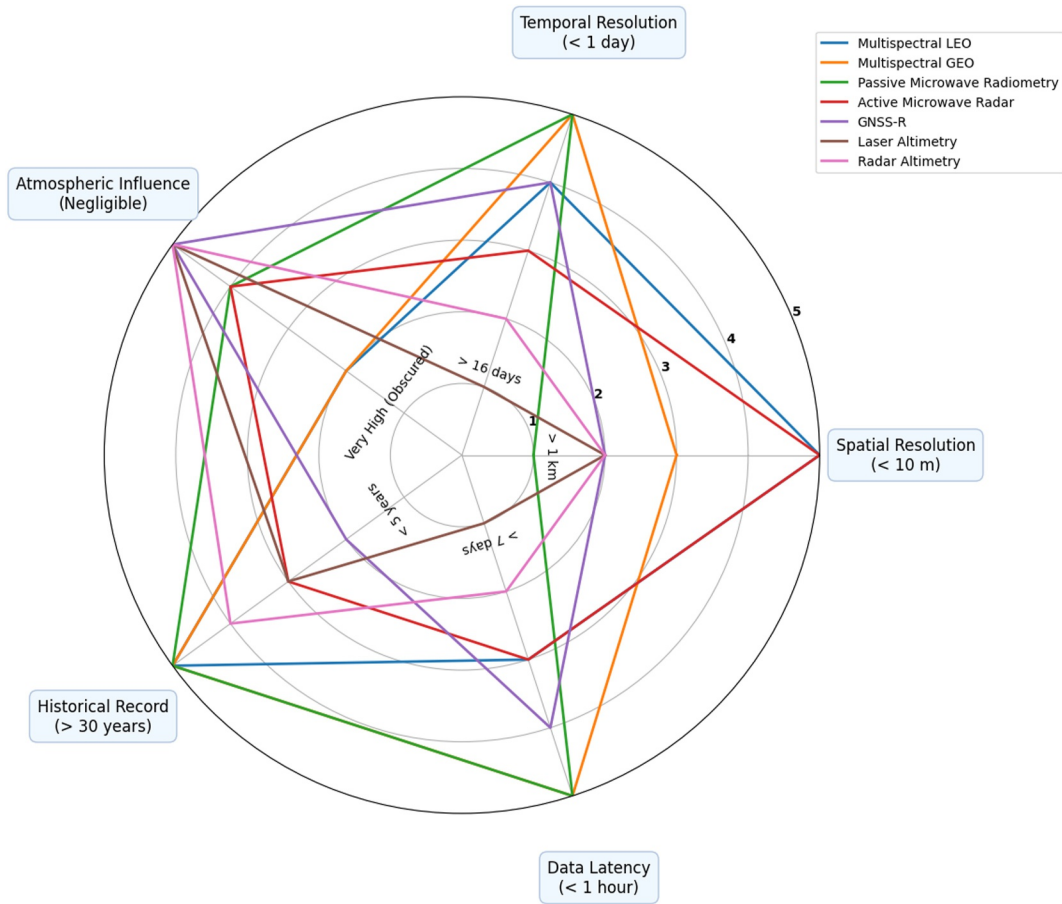



**Figure 3.** Radar chart illustrating the relative strengths and weaknesses of each type of sensor. A higher value (closer to 5) signifies a stronger performance in that area. Please note that this only considers sensors with public, freely available data, as commercial providers may outperform in certain categories.

(Ovando et al., 2018). The synergy between SAR and multispectral LEO sensors best exemplifies this dual benefit of temporal coincidence and physical complementarity. The devastating October 2024 flood event in Valencia, Spain, was captured by both Sentinel-1 and Sentinel-2 satellites on 31 October 2024, providing a multi-sensor view of the event (Figure 5).

The flood extent extracted via SAR appears to underrepresent significantly flooded agricultural and urban areas, whereas these flooded areas in the optical image can be seen clearly. The optical sensor provides the spectral context necessary to identify water in a complex urban environment. Conversely, SAR detected floodwater in areas obscured by clouds, highlighting their utility in tandem. As both acquisitions occurred within the same relevant timeframe, the integration of these physically complementary data sets yields a complete map that neither sensor could produce in isolation. In this context, Markert et al. (2024) made use of these complementary sensor types by applying Sentinel-1 SAR and Landsat-8 optical data to fill gaps in clouded optical imagery, yielding a more consistent time series of surface water dynamics. Similarly, the Observational Products for End-Users from Remote Sensing Analysis (OPERA) project merges SAR and optical information to create detailed surface water maps across diverse environments (Arena et al., 2023).

When adverse atmospheric conditions prevent acquisition of useful optical data, it can be valuable to integrate data from SAR sensors when available, GNSS-R or PMW sensors (Chew et al., 2018; Zheng et al., 2017). Zeng et al. (2020) proposed a hybrid monitoring system that uses PMW sensors to detect floodwater anomalies and SAR sensors for detailed mapping of the area, thus providing more frequent and consistent flood information. A combination between Sentinel-1 SAR and SWOT Ka-band SAR interferometry can provide spatially continuous information of the water surface slope, helping constrain floodplain and channel water exchange (Jaramillo



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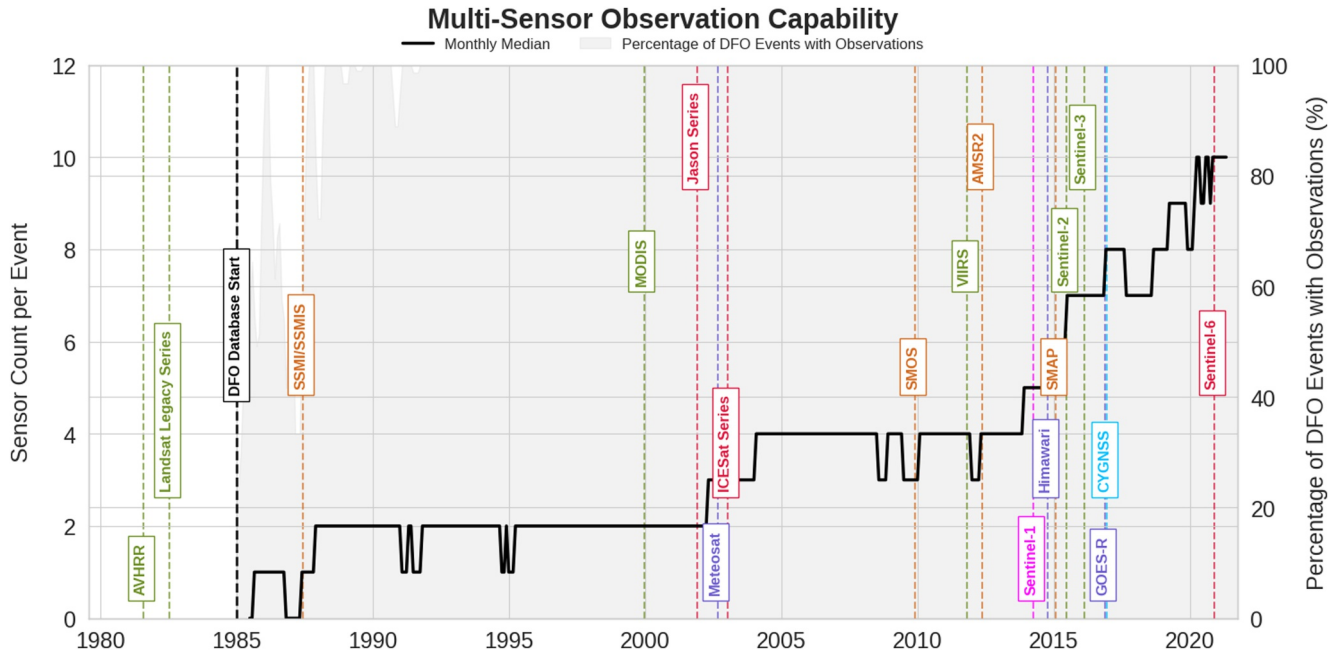



**Figure 4.** The median number of different sensors that have viable data for a single flood event in the DFO database, compiled on a 12-month rolling average. The gray shaded area indicates the percentage of DFO events with observations.

et al., 2024). Finally, data fusion is commonplace in altimetry-based studies. Researchers commonly develop multi-sensor approaches combining data from various altimetric sensors to provide comprehensive time series of water levels over lakes and rivers, providing valuable insights for inundation processes (Boergens et al., 2019; Schwatke et al., 2015; Tourian et al., 2017; Zakharova et al., 2020). The research led by Schwatke et al. (2013) and Tourian et al. (2022) has produced comprehensive time series of water levels, which are readily accessible through DAHITI and HydroSat platforms, respectively.

With the strengths and limitations of the various sensors, it is valuable to consider the temporal inconsistencies that may exist between sensor acquisitions. It is unlikely that different sensors will capture imagery contemporaneously, thus it will always be important to understand how to properly combine information captured at various times. Notti et al. (2018) bridged temporal gaps by combining low-spatial resolution daily satellite data with high resolution SAR imagery from both before and after flooding, using DEM-based models to refine and reconstruct the complete inundation map. To enable flood monitoring with multi-source and multi-temporal imagery, Zhao et al. (2024) developed a deep learning network that performs direct change detection between varied sequences of pre- and post-event optical and SAR images by transferring knowledge from general water extraction models. Furthermore, there may be uncertainties which arise during fusion due to geospatial alignment differences. To remedy this issue, some efforts have been made to harmonize sensor data with similar attributes, such as Sentinel-2 and Landsat, yielding a spectrally consistent and temporally dense data set (Ju et al., 2025). Tulbure et al. (2022) used an earlier version of this harmonized data set to successfully detect ephemeral floods that would have been undetectable by Landsat alone, illustrating the advantage of spatiotemporally dense observations from multiple sensors. Even data sets with variable attributes, both spectrally and spatially, can be aligned to aid in a variety of applications, such as Sentinel-1 and Sentinel-2 (Ye et al., 2021). The emergence of geospatial foundation models has also provided a method to overcome some of these issues, as foundation models using remote sensing data are easily sensor agnostic (Mai et al., 2024).

## 4. Conclusions and Outlooks

Our analysis of the DFO database confirms that there are abundant space-based EO sensors available to capture flooding, indicating that the consistent, large-scale monitoring of floods is well within our reach. The primary challenge is not necessarily the acquisition of data, as there is a tremendous amount publicly available, but rather properly integrating multiple data streams to provide the optimal space-based flood inundation solution. A multi-



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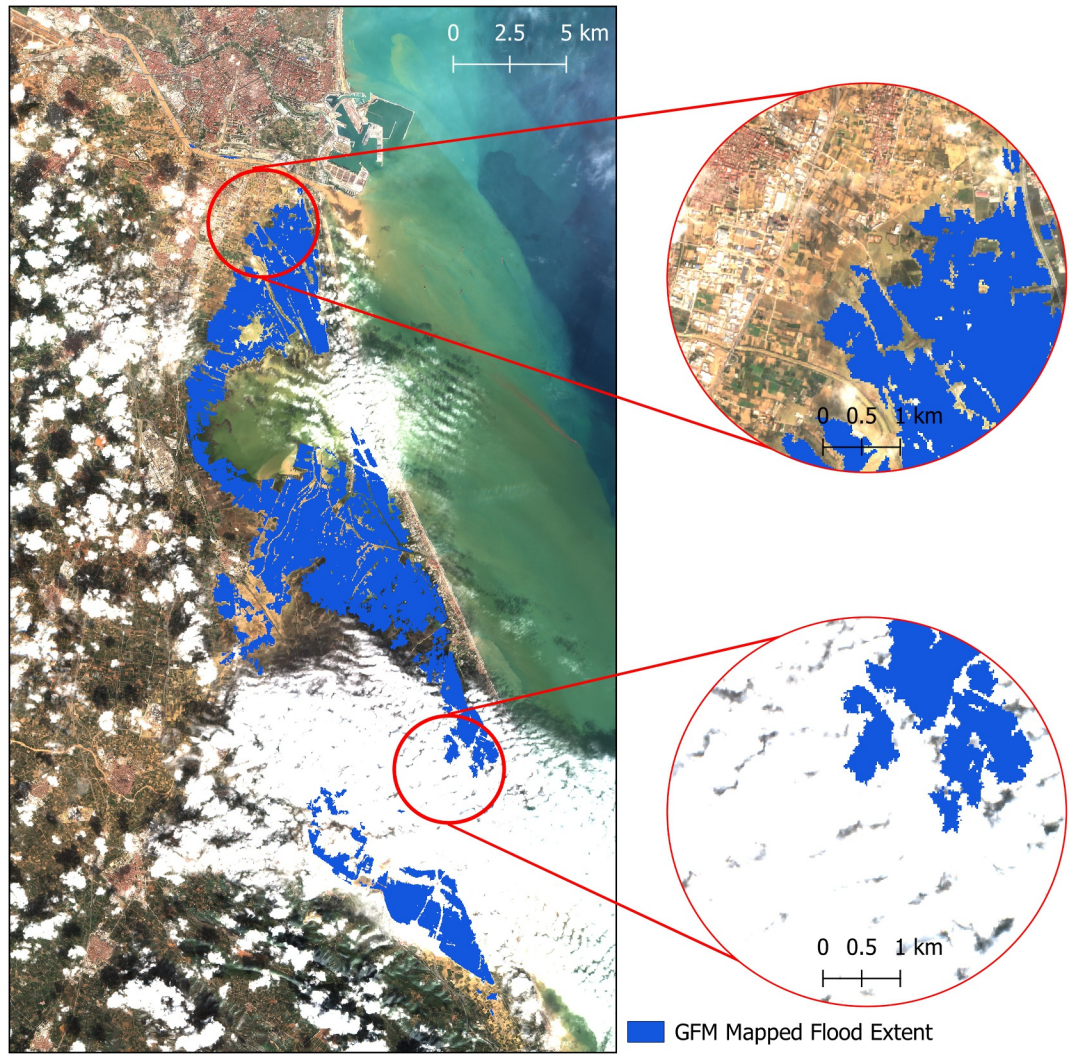



**Figure 5.** Sentinel-2 MSI True Color Image with Sentinel-1 SAR derived flood-extent superimposed on top. The top right circle highlights the missing SAR-derived information, whereas the bottom circle highlights the missing optical information. Sentinel-1 SAR flood extent obtained from the Copernicus Global Flood Monitoring ensemble flood map.

sensor approach, which integrates complementary data, is a promising solution for strengthening our understanding of inundation dynamics. However, optimizing such an approach requires careful consideration of each sensor's operational status, capabilities, and processing requirements.

We believe the future of space-based sensors for hydrology is bright. ESA is continuously expanding its constellations. In the last quarter of 2024, ESA updated the Sentinel-2 constellation, with the new Sentinel-2C. Imagery from Sentinel-2C will be available in preparation for the decommissioning of Sentinel-2A. In the same period, ESA launched Sentinel-1C to replace the defunct Sentinel-1B, cutting the revisit time of this valuable constellation in half. Furthermore, Sentinel-1D was recently launched in November 2025 as a replacement for Sentinel-1A in anticipation of its retirement. ESA will additionally be adding a GNSS-R constellation for global hydrological monitoring, HydroGNSS, in 2025. 2029 will see the launch of ESA's Copernicus Imaging Microwave Radiometer (CIMR) mission, with frequencies ranging from L- to Ka-band. NASA, in collaboration with the Indian Space Research Organization (ISRO), launched the NASA-ISRO SAR (NISAR) mission in July 2025. NISAR will be the first radar imaging satellite to use S- and L-band dual frequencies, making it a monumental resource for future hydrological studies. USGS and NASA are working to launch the Landsat Next constellation in late 2030/early 2031. This constellation will consist of 3 satellites, each equipped with the Landsat Next Instrument Suite (LandIS) which employs 26 bands across visible, shortwave infrared, and thermal infrared ranges,



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providing an improved revisit time of 6 days in comparison to the current 16. This update also includes an improved spatial resolution of up to 10 m. Finally, the AMSR3 was recently launched in June 2025 aboard JAXA's GOSAT-GW mission, serving as an important follow on to AMSR2. These missions are aptly poised to address observational gaps quantified in this review. For instance, the improved 6-day revisit time of LandIS has the possibility to significantly increase the probability of cloud-free observations, addressing the low capture rates of current high-resolution optical sensors. NISAR's L-band radar is specifically designed to penetrate vegetation canopies, with the potential to enhance flood mapping in complex, vegetated floodplains where both optical and C-band SAR sensors are limited.

## Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

## Data Availability Statement

*Software*: The analysis was performed using the Google Earth Engine platform (Gorelick et al., 2017) and the Google Colaboratory environment (Google, 2025).

*Data*: All data used in this study are publicly available from their respective archives. Data for Figure 5 were provided by the Copernicus Emergency Management Service Global Flood Monitoring (GFM) Product Suite (Copernicus Emergency Management Service (CEMS), 2021-present) and the Copernicus Sentinel-2 mission (European Space Agency, 2015) via the Copernicus Data Space Ecosystem.

To determine viable data availability over the locations and durations of flood events, a wide range of satellite remote sensing data sets were assessed. These include products from the Copernicus Data Space Ecosystem (European Space Agency, 2010, 2014, 2015, 2016, 2021); JMA Himawari-8/9 (Japan Meteorological Agency, 2015); the NOAA GOES-R series and AVHRR (NOAA, 2014, 2017, 2018a, 2018b); numerous NASA missions including CYGNSS, Jason, Sentinel-6A, ICESat, ICESat-2, SMAP, AMSR2, SSM/I, SSMIS, and MODIS (CYGNSS, 2017; Desai, 2018; Jackson et al., 2018; JASON-1, 2015; NASA, 2026a, 2026b, 2026c, 2026d, 2026e, 2026f, 2026g, 2026h; Neuenschwander et al., 2023, 2023; ONeill et al., 2021; Vermote & Wolfe, 2021; Zwally et al., 2014); the U.S. Geological Survey Landsat program (Earth Resources Observation and Science (EROS) Center, 1999, 2013, 2020); and EUMETSAT (EUMETSAT, 2020). All data sets are fully cited in the reference list.

The following data sets do not have assigned DOIs and are accessible via the URLs listed below:

- Copernicus Emergency Management Service Global Flood Monitoring (GFM) Product Suite: https://global-flood.emergency.copernicus.eu/
- Copernicus Sentinel-1: https://dataspace.copernicus.eu/data-collections/sentinel-data/sentinel-1
- Copernicus Sentinel-2: https://dataspace.copernicus.eu/data-collections/copernicus-sentinel-missions/sentinel-2
- Copernicus Sentinel-3: https://dataspace.copernicus.eu/explore-data/data-collections/sentinel-data/sentinel-3
- Copernicus SMOS: https://documentation.dataspace.copernicus.eu/Data/ComplementaryData/SMOS.html
- Dartmouth Flood Observatory: https://floodobservatory.colorado.edu/wiki/FloodRecords
- EUMETSAT MSG Cloud Mask: https://user.eumetsat.int/catalogue/EO:EUM:DAT:MSG:CLM
- JMA Himawari-8/9 Standard Data: https://registry.opendata.aws/noaa-himawari/

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